

# Aditya Dasika

Machine Learning Engineer · Multi-Agent Orchestration on LangGraph · Graph RAG & Vector Search · LLM Evaluation, Observability & Serving

[akhil.dasika47@gmail.com](mailto:akhil.dasika47@gmail.com) | [+91-96184-33030](tel:+91-96184-33030) | [Portfolio](#) | [GitHub](#) | [LinkedIn](#)

## EXPERIENCE

### AI/LLM Engineer II

April 2025 – Present

*Phenom People*

*Hyderabad, Telangana*

- Architected **chatbotCXA**gent, a production multi-agent conversational system on **LangGraph** whose orchestration layer coordinates 12 nodes spanning intent recognition, planning, dialogue-state management and tool routing, with **Graph RAG** information retrieval over enterprise knowledge and document repositories via FAISS vector search and a Neo4j graph for multi-hop, policy-aware grounding.
- Engineered tool-calling agents that execute governed actions — database queries and structured extraction from unstructured documents — under **XGrammar constrained decoding** and schema-validated JSON guardrails, raising tool-routing accuracy to **91.9%** and eliminating malformed tool calls in live traffic.
- Built an **LLM evaluation framework** (pass@k, LLM-as-judge rubric) gating every release across all 12 nodes, instrumented with LiteLLM and Langfuse for per-node cost, latency and full-trace auditability — continuously monitoring agent behaviour and curbing hallucinations.
- Fine-tuned and served Qwen models via **QLoRA** (Unsloth, NF4) on **vLLM** / AWS g5.xlarge (A10G), sustaining 10.98 RPS at 98.6% success under load at roughly half the inference cost of hosted GPT-4.1.
- Fine-tuned a multi-task **ModernBERT** (intent + NER) in PyTorch, raising NLU accuracy from 87% to 97%.
- Designed the configuration and policy plane for an enterprise **hiring-integrity / fraud-risk platform**: a global fraud-signal registry with feature groups, tenant-scoped overrides and applicable-context gating, exposed through a **resolve API** consumed by the reviewer console and the live detection pipeline, with a role-gated signal-lifecycle state machine, versioned definitions that keep downstream consumers stable across config changes, and CEL-evaluated decision policies.

### Full Stack Engineer

May 2022 – February 2025

*Blueleaves Farms*

*Hyderabad, Telangana*

- Owned end-to-end delivery of production systems (POS, CRM, inventory, procurement, e-commerce) on Node.js, Firebase Cloud Functions and REST APIs — with role-based access control and CI/CD — integrating multiple internal systems and cutting manual operational errors by roughly 90%.
- Built data-driven analytics on large-scale transactional and stock-movement data — demand forecasting, sales-trend and inventory modeling across 5+ locations and 10,000+ monthly transactions.

## PROJECTS

### IndexNotes — AI Knowledge Graph for Scientific Papers | *Neo4j, Ollama, PyMuPDF, Python, Docker*

- Built an automated Graph RAG pipeline ingesting research papers (PDF to local-LLM extraction to structured JSON to Neo4j), creating 40+ concepts and 17+ typed relationships, with quantized local models via Ollama for entity and relationship extraction.
- Designed a Neo4j schema enabling multi-hop graph traversal and cross-document information retrieval across scientific domains.

### NorthStar — Motorcycle Navigation Companion App | *Android, BLE, Maps SDK*

- Built and shipped a companion app for the Royal Enfield Tripper Dash, delivering turn-by-turn navigation to the bike's connected instrument cluster and growing an organic user base without paid distribution; the traction led to an on-site collaboration visit at Royal Enfield HQ.

## TECHNICAL SKILLS

**Generative AI, LLMs & Agentic Systems:** Agentic / Multi-Agent Systems, LangGraph, LangChain, Tool/Function Calling, Prompt Engineering, Model Routing, Context-Window Management, RAG & Graph RAG, LLM Fine-Tuning (QLoRA, SFT)

**Languages:** Python, JavaScript, TypeScript, SQL, Bash

**Eval, Observability & Guardrails:** LLM Evaluation (LLM-as-judge, pass@k), Constrained Decoding (XGrammar), Schema Validation, Langfuse, LiteLLM, Model Monitoring, Hallucination Mitigation, Role-Based Access Control

**ML, Deep Learning & Serving:** PyTorch, Transformers (BERT/ModernBERT), HuggingFace PEFT, Unsloth, vLLM, Embeddings / Vector Search (FAISS), Neo4j, Ollama, Production MLOps

**Backend & Infra:** FastAPI, Node.js, REST APIs, Microservices, Docker, AWS (EC2/S3/SageMaker), GitHub Actions, CI/CD, Linux

**Databases:** PostgreSQL, Neo4j, MongoDB, Redis, Firebase

## EDUCATION

**Andhra Loyola Institute of Engineering and Technology**

*Bachelor of Technology in Mechanical Engineering*

Vijayawada, Andhra Pradesh

*August 2018 – May 2022*